CUSTOMER CONVERSION INTELLIGENCE PLATFORM

An End-to-End Data Analytics Project

Using Python | SQL Server | Power BI

Analyzing Digital Marketing Campaign Performance, Customer Engagement & Conversion Patterns

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1. Project Overview & Objectives

Project Overview

The Customer Conversion Intelligence Platform is an end-to-end data analytics project focused on analyzing digital marketing campaign performance, customer engagement, and conversion patterns.

The project uses a dataset of 8,000 customer records and 20 attributes. Python, SQL Server, and Power BI were used for data exploration, statistical analysis, structured querying, and interactive visualization.

Project Objectives

- Evaluate conversion performance across different campaign types and marketing channels.
- Analyze customer engagement patterns among converted and non-converted customers.
- Examine conversion patterns across customer segments based on purchase history and loyalty points.
- Explore the relationship between advertising expenditure and conversion.
- Develop an interactive dashboard to present key performance indicators and business insights.
- Translate analytical findings into actionable business observations and recommendations.

2. Business Problem

Digital marketing campaigns generate customer data across multiple channels and touchpoints. However, understanding which campaign categories perform better, how customer engagement relates to conversion, and how customer segments behave requires structured analysis.

The objective of this project is to analyze customer and marketing data to identify conversion patterns, evaluate campaign performance, understand customer engagement, and examine the relationship between advertising expenditure and conversion outcomes.

The analysis focuses on answering three key questions:

1. How do campaign types and marketing channels differ in observed conversion performance?
2. How does customer engagement and purchase history relate to conversion?
3. What insights can be derived from customer segmentation and advertising expenditure?

3. Dataset Description

The project uses a digital marketing campaign dataset containing **8,000 customer records and 20 columns**. Each record represents a customer and includes information related to demographics, marketing campaigns, advertising expenditure, engagement activities, purchase history, loyalty, and conversion.

Dataset Overview

Attribute	Description
Total Records	8,000
Total Columns	20
Data Type	Structured, tabular data
Target Variable	Conversion
Analysis Level	Customer-level

Key Variables

Category	Variables
Customer Demographics	Age, Gender, Income
Campaign Details	CampaignChannel, CampaignType
Advertising	AdSpend, AdvertisingPlatform, AdvertisingTool
Website Engagement	WebsiteVisits, PagesPerVisit, TimeOnSite
Email & Social Engagement	EmailOpens, EmailClicks, SocialShares
Customer History	PreviousPurchases, LoyaltyPoints
Conversion Metrics	ClickThroughRate, ConversionRate, Conversion

Target Variable

The primary outcome variable is **Conversion**, a binary variable:

- 1 — Customer converted.
- 0 — Customer did not convert.

This variable was used to evaluate observed conversion patterns across campaign categories and customer groups.

4. Tools & Technologies

Tool / Technology	Application in the Project
Python	Data cleaning, EDA, feature engineering, statistical analysis, and visualization
Pandas & NumPy	Data manipulation and numerical operations
Matplotlib & Seaborn	Data visualization and exploratory analysis
SQL Server & SSMS	Data querying, aggregation, customer segmentation, and analytical database operations
Power BI	Interactive dashboard development, KPI monitoring, and visual analysis

Key Technical Concepts

- Exploratory Data Analysis (EDA)
- Statistical Hypothesis Testing
- Customer Segmentation
- SQL Joins, CTEs, Subqueries, and Aggregations
- Views, Stored Procedures, and User-Defined Functions
- DAX Measures and Interactive Visualizations

5. Project Methodology

The project followed a structured analytical workflow, progressing from data exploration to business insight generation.

Phase	Activities
1. Data Understanding	Examined dataset structure, variables, and data quality.
2. Python Analysis	Performed data cleaning, EDA, feature engineering, statistical testing, and visualization.
3. SQL Analysis	Conducted structured queries, aggregations, customer segmentation, and reusable database operations.
4. Power BI Development	Created KPI cards, campaign performance charts, customer engagement visuals, segmentation analysis, and interactive filters.
5. Business Interpretation	Consolidated findings, identified limitations, and developed business recommendations.

The workflow combined Python for exploratory and statistical analysis, SQL Server for structured data analysis, and Power BI for interactive business reporting.

6. Python Analysis

Python was used to explore customer data, identify patterns, perform statistical analysis, and generate visual insights.

Data Preparation & Exploration

- Examined dataset structure, data types, and data quality.
- Performed exploratory data analysis using descriptive statistics and visualizations.
- Analyzed customer demographics, campaign attributes, advertising expenditure, website activity, email engagement, and purchase history.

Feature Engineering

Created additional analytical categories to support customer-level comparisons, including:

- Age Groups and Income Categories
- Customer Segments based on Previous Purchases
- Loyalty Categories
- Website Engagement Levels
- Engagement-related metrics

Statistical Analysis

Applied statistical techniques to investigate relationships and differences within the dataset:

- **Chi-square Test:** Examined the association between campaign type and conversion.
- **Independent-Samples t-test:** Compared engagement metrics between converted and non-converted customers.
- **ANOVA:** Examined differences in average pages per visit across campaign types.
- **Correlation Analysis:** Investigated relationships between numerical variables.

Key Python Findings

- Converted customers averaged **9.74 email opens**, compared with **7.58** among non-converted customers.
- Converted customers averaged **7.93 time-on-site units**, compared with **6.27** among non-converted customers.
- The Chi-square test indicated a statistically significant association between campaign type and conversion (p-value approximately 3.398×10^{-18}).
- The t-test identified a statistically significant difference in average pages per visit between conversion groups (p-value approximately 1.09×10^{-17}).
- ANOVA did not identify a statistically significant difference in average pages per visit across campaign types (p-value approximately 0.245).

These results describe patterns and statistical associations within the dataset; they do not establish causal relationships.

7. SQL Analysis

SQL Server was used to perform structured analysis of customer, campaign, engagement, and advertising data. The analysis focused on aggregations, customer segmentation, conversion comparisons, and reusable database operations.

SQL Techniques Applied

- **Joins and Subqueries:** Combined and analyzed related customer and marketing information.
- **Aggregations & Conditional Logic:** Calculated customer counts, conversion rates, averages, and campaign-level metrics.
- **CTEs:** Organized complex analytical queries into logical steps.
- **Window Functions:** Applied analytical ranking and comparison techniques where required.
- **Views:** Created reusable analytical views for customer summaries, campaign performance, advertising platform analysis, loyalty, and engagement.
- **Stored Procedures:** Developed reusable procedures for customer summaries, campaign performance, advertising platform analysis, and loyalty reports.
- **User-Defined Functions:** Created functions for engagement scoring, loyalty categorization, income categorization, and customer segmentation.

Key SQL Findings

- Analyzed **8,000 customer records**, identifying 7,012 converted customers and an overall conversion rate of 87.65%.
- The Conversion campaign category recorded an observed conversion rate of approximately **93.36%**.
- Customers in the Regular segment had an observed conversion rate of approximately **90.95%**, compared with 77.33% among New customers.
- Customers in the Very High loyalty category recorded an observed conversion rate of approximately **90.20%**, compared with 78.95% in the Low loyalty category.
- Customers with above-average advertising spend showed a higher observed conversion rate than those with average or below-average spend.

These findings highlight differences across campaign and customer groups. They should be interpreted as descriptive associations rather than evidence of causal impact.

8. Power BI Dashboard

An interactive, single-page Power BI dashboard was developed to present key performance indicators and explore campaign performance, customer engagement, and customer conversion patterns.

Key Performance Indicators (KPIs)

The dashboard includes four primary KPIs:

- **Total Customers:** 8,000
- **Converted Customers:** 7,012

- **Overall Conversion Rate:** 87.65%
- **Average Ad Spend:** 5,000.94

Dashboard Visualizations

Six analytical visuals were developed across three business areas:

Business Area	Visualization
Campaign Performance	Conversion Rate by Campaign Type
Campaign Performance	Conversion Rate by Campaign Channel
Customer Engagement	Average Pages per Visit by Conversion Status
Customer Engagement	Average Email Opens by Conversion Status
Customer Profile	Conversion Rate by Customer Segment
Customer Profile	Conversion Rate by Loyalty Category

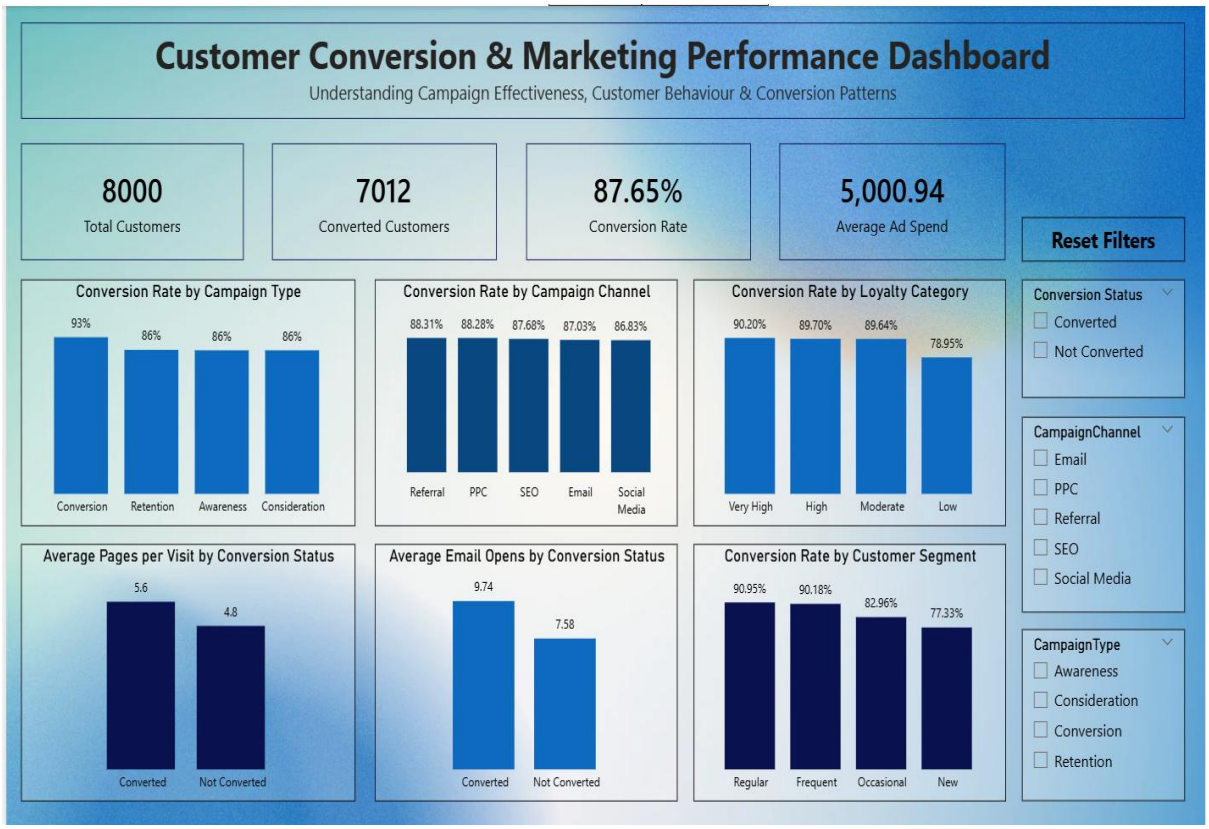


Figure 1: Customer Conversion & Marketing Performance Dashboard

Interactive Features

- Campaign Type, Campaign Channel, and Conversion Status slicers for interactive filtering.
- Reset Filters button to restore the default dashboard state.
- KPI cards and charts designed to support visual comparison and exploration.

- The dashboard provides a consolidated view of marketing performance and customer behaviour, enabling users to explore conversion patterns across different campaign and customer groups.

9. Key Findings & Business Insights

The analysis identified the following key patterns in campaign performance and customer behaviour:

Campaign Performance

- The Conversion campaign category recorded the highest observed conversion rate at **93.36%**, compared with approximately 85.56% for Awareness and Consideration campaigns.
- Referral had the highest channel conversion rate at **88.31%**, although differences across channels were relatively small.

Customer Engagement

- Converted customers showed higher average email opens (**9.74 vs. 7.58**) and pages per visit (**5.6 vs. 4.8**) than non-converted customers.
- Average time on site was also higher among converted customers (**7.93 vs. 6.27**).

Customer Profile

- Regular customers recorded a higher observed conversion rate (**90.95%**) than New customers (**77.33%**).
- Customers in the Very High loyalty category recorded a conversion rate of **90.20%**, compared with **78.95%** in the Low loyalty category.

Advertising Spend

- Customers with above-average advertising spend showed a higher observed conversion rate (**92.30%**) than customers with average or below-average spend (**82.98%**).

Overall, campaign category, customer engagement, purchase history, and loyalty showed notable associations with conversion outcomes. However, these patterns do not establish direct causal relationships.

10. Business Recommendations & Limitations

Business Recommendations

Based on the observed patterns, the following recommendations can be considered:

1. **Campaign Evaluation:** Investigate the higher observed conversion rate of the Conversion campaign category and validate its performance before making budget allocation decisions.
2. **Customer Engagement:** Explore ways to improve website browsing and email engagement, particularly among non-converted customers.
3. **Customer Retention:** Examine strategies to engage new and low-loyalty customers, as these groups showed comparatively lower conversion rates.
4. **Advertising Spend:** Evaluate advertising expenditure alongside revenue, profitability, and conversion outcomes before assessing campaign efficiency.

Project Limitations

- The dataset is synthetic, so the findings may not represent actual marketing behaviour.
- The overall conversion rate of 87.65% is unusually high and should be interpreted cautiously.
- Advertising platform and advertising tool columns each contain only one distinct value, limiting comparative analysis.
- Revenue and profit information is unavailable, preventing reliable ROI and profitability calculations.
- The analysis identifies associations and differences but does not establish causal relationships.
- Some differences across customer demographics and marketing channels are relatively small.

These limitations should be considered before applying the findings to real-world marketing decisions.

11. Conclusion

The Customer Conversion Intelligence Platform demonstrates the application of Python, SQL Server, and Power BI to analyze digital marketing performance and customer conversion patterns.

Through exploratory data analysis, statistical testing, structured SQL analysis, and interactive visualization, the project identified notable patterns in campaign performance, customer engagement, purchase history, and loyalty.

The project provided practical experience in transforming raw data into meaningful business insights, interpreting analytical results, and communicating findings through data visualization.

Overall, this project strengthened the understanding of how data analytics can support marketing performance evaluation and informed business decision-making.